

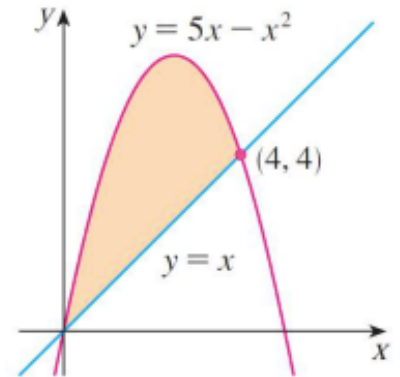
Name _____

Date _____

Total
/32

1. Determine the area of the shaded region in the diagram below to the nearest tenth.

(3)



2. Determine the exact area of the region enclosed by the following curves.

$$y = x + 1, \quad y = (x - 1)^2, \quad x = -1, \quad x = 2$$

(4)

3. Determine the exact area of the region enclosed by the following curves.

$$y^2 = x, \quad x - 2y = 3$$

(3)

4. A particle is moving along the x-axis with velocity $v(t) = 4 - t^2$, where velocity is measured in m/s. Its initial position is $s(0) = 5m$.

a. Describe the particle's position after the first 3 seconds.

(2)

b. Determine to the nearest tenth of a meter, the total distance travelled in the first 7 seconds.

(3)

5. The region enclosed by the curves below is rotated about the x-axis. Determine, to the nearest tenth, the volume of the solid generated. Include a diagram.

$$y = \frac{1}{x}, \quad x = 1, \quad x = 2, \quad y = 0$$

(3)

6. The integral $\int_0^{\frac{\pi}{2}} 2\pi \cos x dx$ represents the volume of a solid. Describe the solid.

(2)

7. Otto was determining the volume of the solid formed by revolving the region enclosed by $y = 1$ and $y = \sin x + 1$, where $0 \leq x \leq \pi$, about the line $y = -2$. To do so, he used the washer method and arrived at the following definite integral. Is Otto's integral correct? Explain.

$$V = \pi \int_0^{\pi} [(\sin x + 1)^2 - 1^2] dx$$

(3)

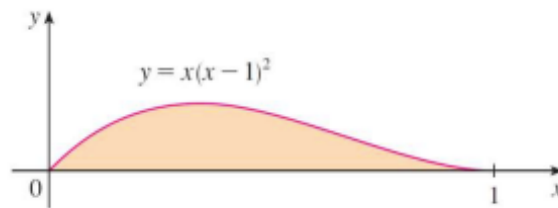
8. The region enclosed by the curves below is rotated about the y-axis. Determine the exact volume of the solid generated. Include a diagram.

$$y = \sqrt{x}, \quad y = \frac{x}{2}$$

(4)

9. The region shown on the right is rotated about the y-axis. Use the Shell method, to determine the exact volume of the region generated.

(3)



10. Set up, but do not evaluate, an integral for the volume of the solid obtained by rotating the region bounded by the given curves about the specified axis.

$$y = \cos^2 x, \quad |x| \leq \frac{\pi}{2}, \quad y = \frac{1}{4}, \quad \text{about } x = \frac{\pi}{2}$$

(2)